MIL-S-60140A(MU)
12 August 1966
SUPERSEDING
MIL-S-60140(MU)
18 January 1965

MILITARY SPECIFICATION

SIGHTING SYSTEM ANTIOSCILLATION: XM58

1. SCOPE

1.1 This specification covers one type of antioscillation sighting system employed in a UH-1B helicopter. The sighting system compensates for vibrations and attitude changes of the helicopter (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic-and Electrical Component Parts

DRAWINGS

U.S. Army, Frankford Arsenal

F10527235	Sighting System, Antioscillation: 10527235

FSC 1270

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PACKAGING DATA SHEET

MIL-P-14232/P10527235 Packaging of Sighting System, Anti-oscillation: **10527235**

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article testing.- Requirements for the submission of first article samples by the contractor for testing shall be as specified in the contract.

3.2 Fabrication. - The sighting system shall be fabricated in accordance with Drawing F10527235 and drawings, specifications and all standards pertaining thereto.

3.3 General specification.- The contractor shall be responsible for adherence to the following requirements of MIL-F-13926:

a. Dimensions and tolerances
b. Inorganic protective surface finishes
c. Part identification and marking
d. Workmanship
e. Soldering electrical connections

3.4 Performance.

3.4.1 Image angular motion.- The angular motion of the line of sight through the telescope shall not exceed 20 seconds of arc peak to peak when subjected to the following 'Vibrational inputs along three mutually perpendicular axes:

0.10" Double amplitude 2 5.5 cps
0.125" Double amplitude @ 11.0 cps
.7g at 22 cps

3.4.1.1 A target will be used to determine the performance and acceptance of the production sights. The amount of angular motion of the line of sight through the telescope shall not exceed sixty seconds of arc peak to peak 90% of the time when the operator utilizes a field of view confined to 2 degrees of the optical center of the monocular. The target shall be so designed that, when observed at a range of 500 meters, more than 60 seconds of arc peak to peak will result in blurring of the image.

3.4.2 System dynamic characteristics.- The time response of each channel from time of step-race application to the peak of the first overshoot shall be 0.09 second or less. The time required for each stabilization servo transient response to settle to within 5% of its final value shall be 0.4 second. The peak overshoot shall not exceed 60% of its total excursion between time zero. and its final value.

2. –"———————————— — --—..

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3.4.3 Power requirements.- The sighting system shall operate with a nominal input voltage of 28 volts direct current (VDC). having an average power consumption of 50 watts and peak power consumption of 150 watts.

3.5 Radio interference suppression--- Interference control requirements shall be in accordance with MIL-I-6181.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract *or* purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article testing.- When the contract specifies testing of preproduction or initial production samples, the following paragraphs shall apply as applicable.

4.2.1 Preproduction test.- Preproduction test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3, 4.4 and 4.6 as well *as* any additional tests that may be specified in the contract.

4.2.2 Initial production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4 and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

4.2.3 First article disapproval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as required by the contract clauses applicable to first article testing.

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4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contain in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (lEL), when provided with the technical data package (TDP). In the absence of SQAP'S the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I and herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I and II shall constitute the **minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.**

TABLE I

(100 Percent Inspection)

CHARACTERISTICS	REQUIREMENT	TEST PROCEDURE
101. Image angular motion	3.4.1	4.6.1.1
102. System dynamic characteristics	3.4.2	4.6.1.2
103. Power requirement	3.4.3	4.6.1.3

MINOR: **NONE**

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 **Special sampling.- Three systems shall be selected at random as a special sampling from each 100** produced. The sample shall meet the requirements and tests in Table II.

IABLE II

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
401. Fabrication	3.2	Applicable drawing visual
402. **General specification**	**3.3**	**MIL-F-13926 - visual**
403. Radio interference suppression	**3.6**	**4.6.2** "

. 4.

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4.4.1 Failure of sample. - Should one item of :.-. al sampl - ing fail to meet the specified test requirements, accept a?.:-;=. of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection eqiupment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus , concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Performance.

4.6.1.1 Image angular motion.- With the system energized and functioning normally, i.e., line-of-sight through the monocular gyro stabilized, sight on the reticle of an autocollimator. Adjust the distance from the collimator to the sight so that a clear image results. Peak to peak deviation of the reticle image shall be less than 20 seconds of arc when the assembly is vibrated according to the schedule in 3.4.1. A target consisting of three six inch wide vertical black bars spaced six inches apart and three similar horizontal lines all painted on a 4 ft. by 8 ft. plywood panel with a white background is to be mounted in an obstructed area where it can be seen from a UH-lB helicopter hovering 500 meters away. The target is to be sighted through the sight. If the peak to peak vibration exceeds 60 seconds of arc,

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the black and white lines will be superimposed on each other causing the target to look gray or to start showing a fourth line.

4.6.1.2 System dynamic characteristics.- Mount the sight assembly in a holding device in its normal attitude. Wire in the test setup shown in Figure 1. Connect all required electronic hardware, i.e.; electronic control amplifier, gimbal position control, etc. Provide required power to the system. Connect an oscillographic recorder across each torquer. Place the "azimuth - elevation" switch in "azimuth". Manually position the elevation gimbal to be orthogonal to the azimuth gimbal. Set the d.c. supply for 15 volts. Put the "operate - test" switch in "test" and then return to "operate" to observe the azimuth channel dynamic response characteristics. This test should be done in both the "normal" and "reverse" modes. Note that when the "operate - test" switch is in the "test" position the azimuth gimbal will turn at a constant rate. Care must be taken to insure that the test is not attempted with the gimbal against its stops. Repeat this test again for the elevation channel. In this case set the d.c. supply for 10 volts. The step response of each gimbal servo shall have a waveform as shown in Figure 2.

4.6.1.3 Power requirement.- Using standard power measurement equipment, measure the power consumption of the energized system for quiescent operation (no external disturbances to gimbals or gyros) and for dynamic operation (both gimbals being maximum torqued). Power consumption for quiescent and peak operation shall coincide with the average and peak power requirements of 3.4.3.

4.6.2 Radio interference test.- The equipment shall be tested in accordance with MIL-I-6181 to ensure conformance with 3.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10527235. The level of protection shall be as specified in the procurement document.

5.2 Technical manuals.- Unless otherwise stated in the procurement document, two copies of applicable technical manuals will be furnished by the Government for pack up and shipment with each sighting system.

6. NOTES

6.1 Intended use.- The Antioscillation Sighting System XM58 employed in helicopters is intended to aid the operator of the helicopter mounted guided missile system to:

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(a) acquire a target through *use of* a six-power, twelve-degree field of view telescope mounted in the gyro-stabilized sight assembly.

(b) stabilize the gimbals of the sight assembly dynamically in inertial space so the line of sight will be sufficiently stabilized to keep the target and missile within the field of view during flight.

(c) indicate to the pilot by LOS Indicator when the ship is pointed along the line of sight of the optics.

6.2 ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing (see 5.1).
(c) Applicable packaging data sheet number (see 5.1).
(d) A statement regarding pack up requirements for technical manuals (see 5.2).

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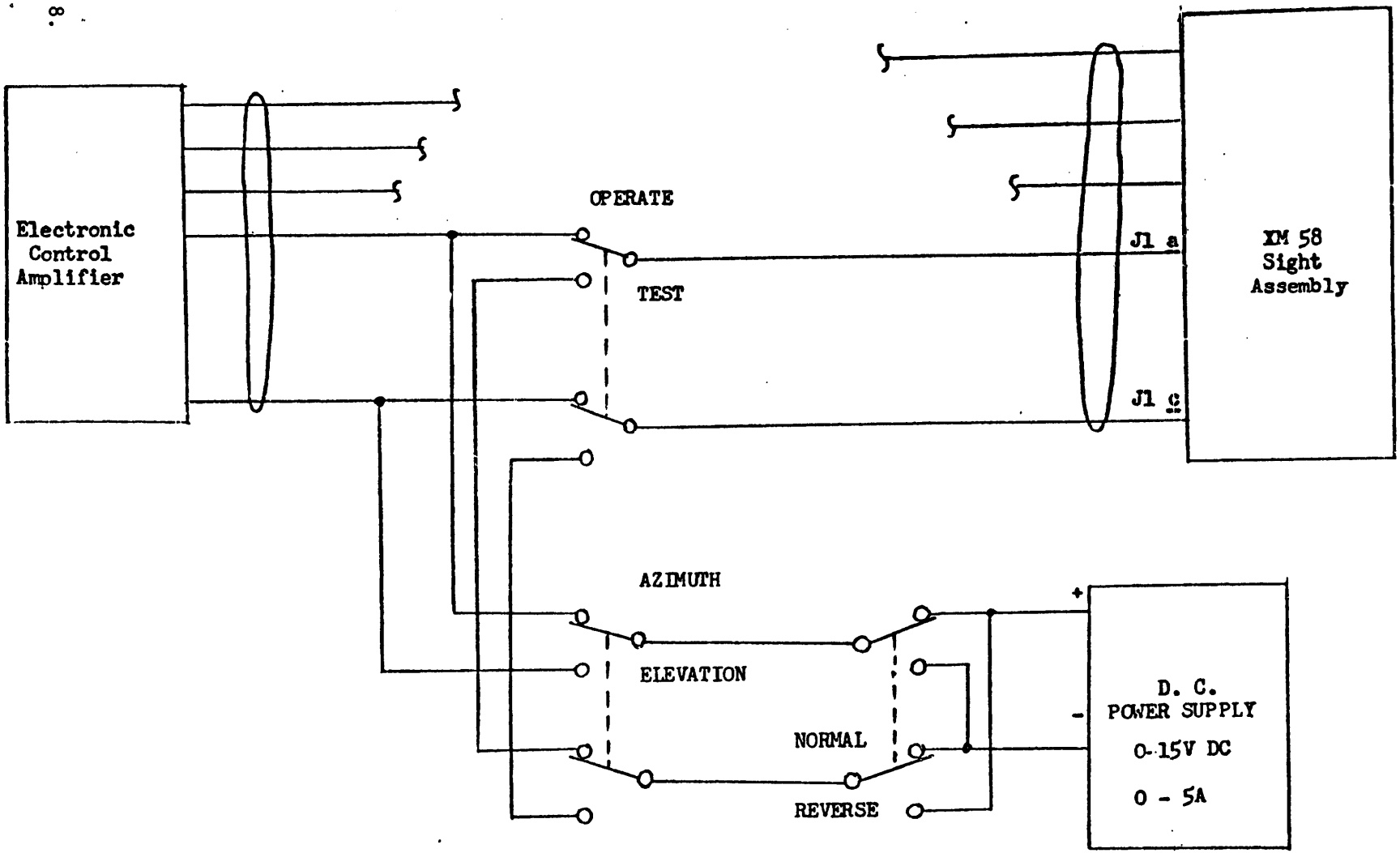


FIGURE #1

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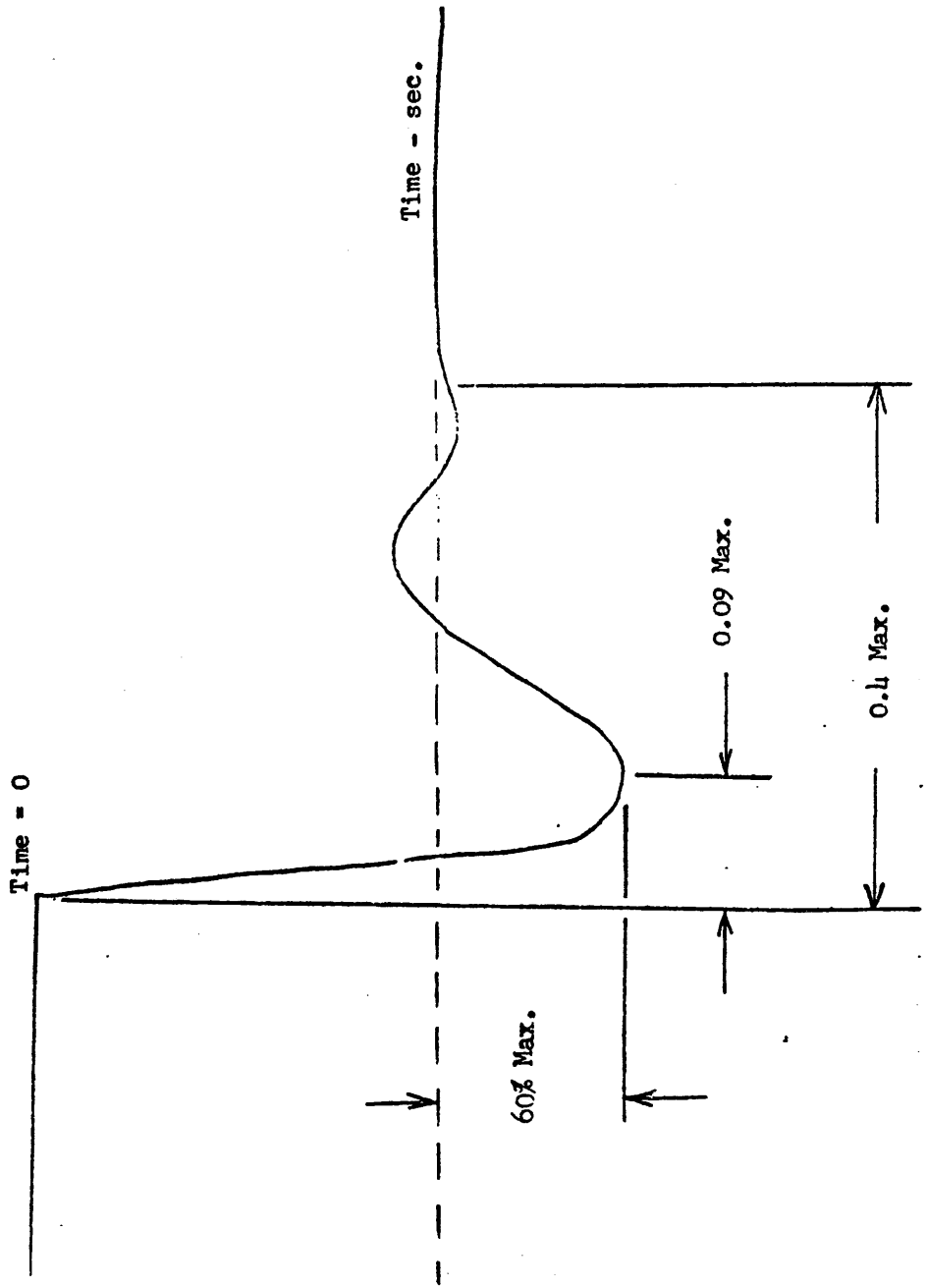


FIGURE #2

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R094

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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